

OSIJEK KLISA, Croatia

WMO: 142840

Lat: 45.468N

Lon: 18.806E

Elev: 89

StdP: 100.26

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-7.9	-14.0	1.1	-9.6	-10.8	1.5	-6.4	10.3	3.2	9.2	3.3	2.7	340	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	34.3	21.1	32.9	20.8	31.1	20.1	23.1	29.4	22.3	29.1	21.5	28.3	3.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.0	25.7	20.2	15.0	24.9	19.2	14.1	23.8	68.9	30.0	65.9	29.3	62.9	28.4	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.8	6.7	-15.3	36.9	6.0	2.3	-19.6	38.5	-23.1	39.8	-26.5	41.1	-30.8	42.7		
(5)			-15.5	24.7	5.9	1.3	-19.7	25.7	-23.2	26.4	-26.5	27.2	-30.7	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	1.1	2.5	7.4	13.2	17.3	21.1	22.9	22.6	17.7	12.9	7.2	2.1		
	DBStd	8.76	4.82	5.16	4.66	3.56	3.71	3.58	3.11	3.11	3.58	3.99	4.19	4.36		
	HDD10.0	974	277	211	103	13	2	0	0	0	0	19	102	247		
	HDD18.3	2622	535	443	339	157	64	15	3	3	53	172	333	504		
	CDD10.0	1842	1	1	22	108	227	333	399	391	231	108	19	2		
	CDD18.3	448	0	0	0	3	31	98	144	136	33	3	0	0		
	CDH23.3	4974	0	0	1	74	349	1047	1592	1483	374	53	0	0		
	CDH26.7	2024	0	0	0	13	87	405	724	678	113	4	0	0		
(14) Wind	WSAvg	3.1	3.2	3.4	3.7	3.5	3.4	2.9	2.8	2.6	2.7	2.7	3.1	3.2		
(15) Precipitation	PrecAvg	678	43	46	45	51	82	73	68	60	56	60	52	47		
	PrecMax	951	77	83	88	131	181	143	185	200	126	138	116	110		
	PrecMin	396	12	4	4	4	25	28	21	4	3	2	4	2		
	PrecStd	152	18	21	24	33	43	30	42	47	32	35	25	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.8	22.8	28.7	31.2	34.2	37.8	37.2	32.9	27.2	21.0	14.9		
		MCWB	10.1	10.2	12.8	16.9	18.8	21.3	21.2	21.9	18.9	17.0	14.3	10.7		
	2%	DB	11.2	14.0	20.1	25.2	29.0	32.8	34.7	34.8	29.9	24.9	17.9	12.1		
		MCWB	8.1	9.6	11.8	15.1	17.9	21.3	21.2	21.5	18.1	16.0	13.0	9.8		
	5%	DB	9.0	11.2	17.8	23.1	27.1	30.9	32.9	32.4	27.2	22.9	15.2	10.0		
		MCWB	7.2	7.8	10.5	13.7	17.4	20.4	21.0	20.6	17.3	15.2	12.2	8.2		
	10%	DB	7.1	9.1	15.1	21.0	25.1	29.0	30.9	30.7	25.2	20.2	13.3	8.0		
		MCWB	5.8	7.0	9.4	12.8	16.5	19.9	20.4	20.0	16.6	14.5	10.9	6.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	11.3	13.6	17.6	20.5	24.0	23.8	24.3	20.7	18.3	15.5	11.5		
		MCDB	13.2	15.2	19.6	25.6	27.0	29.5	29.6	31.4	25.8	23.6	19.2	13.9		
	2%	WB	8.9	9.7	12.4	16.0	19.4	22.9	22.9	22.9	19.4	16.9	13.9	10.0		
		MCDB	11.0	13.1	18.2	23.4	25.7	28.5	29.5	29.6	25.8	22.5	16.6	11.9		
	5%	WB	7.1	8.4	11.3	14.8	18.4	21.9	22.2	22.0	18.6	16.0	12.6	8.4		
		MCDB	8.4	10.7	16.3	20.6	24.5	28.2	29.4	29.3	24.8	21.2	15.2	10.0		
	10%	WB	5.7	7.1	10.1	13.7	17.5	21.0	21.3	21.1	17.8	15.1	11.2	6.6		
		MCDB	6.8	8.8	14.4	19.2	23.4	27.3	28.3	28.0	23.4	19.5	13.5	7.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.9	10.5	12.1	11.4	11.8	12.8	13.0	11.6	11.2	8.2	5.9		
		MCDBR	9.9	12.1	15.4	16.3	14.7	14.7	16.1	16.6	15.7	15.2	11.5	9.3		
	5% WB	MCWBR	7.0	7.7	7.9	7.3	5.4	5.4	4.9	5.3	5.7	7.2	7.2	6.9		
		MCWBR	9.0	10.6	13.2	13.9	13.0	13.2	13.9	14.4	12.9	13.6	10.5	8.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.341	0.366	0.399	0.455	0.441	0.439	0.443	0.447	0.430	0.396	0.361	0.341		
		taud	2.362	2.284	2.228	2.080	2.178	2.224	2.207	2.203	2.219	2.311	2.362	2.366		
	Ebn at Noon	Ebn at Noon	751	798	822	804	831	833	823	804	781	754	717	706		
		Edh at Noon	77	100	121	153	143	138	138	134	121	97	77	70		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.22	2.03	3.29	4.61	5.62	6.16	6.22	5.54	3.84	2.63	1.52	0.98		
	RadStd	RadStd	0.19	0.29	0.41	0.43	0.48	0.50	0.46	0.55	0.43	0.27	0.14	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	+0.67	N/A	N/A	N/A	+0.42	+0.44	-117	-183	+114	+61

Nomenclature: See separate page